Medicine & Science IN Sports & Exercise
The Official Journal of the American College of Sports Medicine
www.acsm-msse.org

. . . Published ahead of Print

Effects of Repetitive Head Impacts on a Concussion Assessment Battery

Jaclyn B. Caccese[1], Chelsea Best[2], Lindsey C. Lamond[3], Melissa DiFabio[1], Thomas W. Kaminski[1], Dan Watson[4], Nancy Getchell[1], and Thomas A. Buckley[1]

[1]University of Delaware, Department of Kinesiology and Applied Physiology and Interdisciplinary Biomechanics and Movement Science Program, Newark, DE; [2]Indiana University Athletics, Bloomington, IN; [3]Wake Forest Baptist Health, Department of Family Medicine, Winston-Salem, NC; [4]University of Delaware Athletics, Newark, DE

Accepted for Publication: 9 January 2019

ACCEPTED

***Medicine & Science in Sports & Exercise*® Published ahead of Print** contains articles in unedited manuscript form that have been peer reviewed and accepted for publication. This manuscript will undergo copyediting, page composition, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered that could affect the content.

Copyright © 2019 American College of Sports Medicine

Medicine & Science in Sports & Exercise, Publish Ahead of Print
DOI: 10.1249/MSS.0000000000001905

Effects of Repetitive Head Impacts on a Concussion Assessment Battery

Jaclyn B. Caccese[1], Chelsea Best[2], Lindsey C. Lamond[3], Melissa DiFabio[1], Thomas W. Kaminski[1], Dan Watson[4], Nancy Getchell[1], and Thomas A. Buckley[1]

[1]University of Delaware, Department of Kinesiology and Applied Physiology and Interdisciplinary Biomechanics and Movement Science Program, Newark, DE; [2]Indiana University Athletics, Bloomington, IN; [3]Wake Forest Baptist Health, Department of Family Medicine, Winston-Salem, NC; [4]University of Delaware Athletics, Newark, DE

Corresponding Author:

Thomas A. Buckley

100 Discovery Blvd, Newark, DE 19713

Telephone: 302-831-4783

Email: tbuckley@udel.edu

This publication was made possible, in part, with support from the Grand Alliance CARE Consortium, funded by the NCAA and the DoD. The USAMRAA, Ford Detrick, MD, USA, is the awarding and administering acquisition office. This work was supported by the Office of the Assistant Secretary of Defense for Health Affairs through the Psychological Health and Traumatic Brain Injury Program under Award No. W81XWH-14-2-0151. Opinions, interpretations, conclusions and recommendations are those of the authors and are not necessarily endorsed by the DoD (DHP funds).

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Conflict of Interests: The authors have no conflicts of interest to disclose. The results of the present study do not constitute endorsement by ACSM. The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation.

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Abstract

Purpose: The purpose of this study was to determine the relationship between repetitive head impacts (RHI) and clinical concussion assessments across a season among collegiate football (FB) and women's soccer (WSOC) players. **Methods:** Fifteen male FB and 23 WSOC players participated in this study. Participants were included if they were medically cleared for unrestricted athletic participation. Participants were tested in a university athletic training room on two occasions: pre-season (PRE) and post-season (POST). The outcome measures consisted of Tandem Gait (TG), Standardized Assessment of Concussion (SAC), Balance Error Scoring System (BESS), King-Devick (KD), Clinical Reaction Time (CRT), and Immediate Post-Concussion Assessment and Cognitive Testing (ImPACT). RHI during the season was quantified using the Head Impact Telemetry System (HITS; Simbex, NH) for FB and the Smart Impact Monitor (SIM; Triax Technologies, CT) for WSOC. Independent variables included total number of impacts, average magnitude of peak linear acceleration, cumulative linear exposure, and number of impacts $\geq$98g. **Results:** Results from direct-entry multiple regression analyses suggest significant associations between RHI and both Visual Memory (R^2=0.670, F=6.487, p=0.002) and TG (R^2=0.636, F=3.841, p=0.029) for WSOC and between RHI and KD (R^2=0.756, F=5.579, p=0.013) for FB, whereby those with greater exposure performed worse. No other regression analyses within or across groups were significant. **Conclusions:** These data suggest that RHI do not represent clinically meaningful changes on a multifaceted and multimodal concussion assessment battery. However, there may be subtle visual/vestibular impairments as observed by the associations between RHI and Visual Memory/TG among WSOC, RHI and KD among FB.

Key Words: soccer, football, subconcussion, mTBI, head acceleration

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Introduction

Repetitive head impacts (RHI) have been defined as mild impacts that do not result in a known or diagnosed concussion on clinical grounds (1). There is growing public concern regarding the potentially adverse effects of RHI through routine participation in contact and collision sports (e.g. football, soccer, hockey and rugby). Frequent exposure to RHI have been associated with long-term neurological impairments, including functional deficits, blood brain barrier dysfunction, abnormal neuro-metabolic and neuro-inflammatory processes, as well as tau aggregation (2-4). Contrarily, other studies have suggested that there are no neurological impairments due to RHI exposure in sports (5-7). Given the frequency of head impacts in contact and collision sports, the public health implications of these consequences may be significant.

While the long-term effects of RHI have been studied across a number of sports, including football, ice hockey, and soccer (2-4), the short-to-medium-term (i.e. across one season of participation) effects of RHI have been relatively understudied and largely limited to high school and collegiate football (FB) and ice hockey players (8-12). The studies on high school and collegiate FB suggest that RHI exposure has little effect on clinical measures of neurological impairment (e.g., neurocognitive testing), despite subclinical changes, such as altered functional Magnetic Resonance Imaging (fMRI) and diffusion tensor imaging (DTI) (8-12). Yet, there is a lack of research on the effects of RHI in soccer using objective methods for quantifying RHI.

Both researchers and funding agencies have been addressing the historic paucity of attention over sex influences on recovery outcomes after concussion and RHI (13). Moreover, female athletes may be more susceptible to structural alterations following exposure to RHI than male athletes (14). Therefore, it is imperative to include female participants in studies evaluating the effects of RHI in sport. According to the 2016-2017 NCAA Sports Sponsorship and

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Participation Rates Report, women's soccer (WSOC) has the second highest number of participants and RHI among WSOC is frequent, although not as frequent as in FB (i.e. ~100 per season in WSOC vs. ~1000 per season in FB) (8, 15). There have been mixed results regarding the clinical and subclinical effects of RHI in collegiate WSOC on the field (16). However, previous work did not objectively quantify both number and magnitude of RHI sustained by WSOC in their sample, thus the potential dose-response of RHI and neurological changes in unknown in collegiate WSOC. Adult recreational soccer players appear to have a RHI dose-response whereby increased soccer heading was associated with poorer performance on neurocognitive tests and abnormal white matter microstructure (17, 18). Yet, no study has previously explored the relationship between RHI and clinical measures of neurologic health while monitoring sustained impacts throughout the course of a single collegiate WSOC season.

Multifaceted and multimodal assessment batteries are sensitive in acute concussion diagnosis, and often include computerized neuropsychological testing, postural control assessments, and clinical reaction time testing (19, 20). However, a multifaceted and multimodal battery has never been used to assess changes in neurological health from pre-season to post-season in a cohort of both collegiate FB and WSOC athletes utilizing common neurological screening assessment tools simultaneously. Using this battery of assessment tools may be more sensitive to changes from RHI than using one clinical outcome measure, such as the ImPACT test, alone. Therefore, the purpose of this study was to determine the relationship between RHI on clinical concussion assessments across a season of collegiate FB and WSOC. Potential contributors of neurologic deficits according to frequency and magnitude of impacts were all considered in the analyses. We hypothesized that RHI over a single season would not result in neurologic impairment among our FB cohort but would result in neurologic impairment among

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

our WSOC cohort considering that females may be more susceptible to neurological alterations following exposure to RHI (11).

Methods

Participants

This study recruited 38 National Collegiate Athletic Association (NCAA) Division-I student-athletes including 15 FB players and 23 WSOC players (Table 1). Participants were included if they were medically cleared for full, unrestricted athletic participation on the day of testing and had not suffered a concussion within the previous 60 days of the pre-season test and did not suffer a concussion during the season. Participants who sustained musculoskeletal injuries but were still listed as full participation at the end of the season were included in the study. The exclusion criteria included any condition that restricted medical clearance at either time of testing. All participants provided oral and written informed consent as approved by the University of Michigan IRB, the US Army Medical Research and Materiel Command Human Research Protection Office (HRPO), as well the local IRB at the university.

Study Design

Participants were tested on two occasions: 1) pre-season (PRE), during the summer within two weeks prior to the start of their competitive season and 2) post-season (POST), within one week after the end of their competitive season. Testing consisted of the Tandem Gait (TG), Standardized Assessment of Concussion (SAC), Balance Error Scoring System (BESS), King-Devick (KD), Clinical Reaction Time (CRT), and Immediate Post-Concussion Assessment and Cognitive Testing (ImPACT). Exposure to RHI was quantified using the Head Impact Telemetry System (HITS; Simbex, Lebanaon, NH, USA) for FB and the Smart Impact Monitor (SIM;

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Norwalk, CT, USA) for WSOC. For both accelerometer systems, the magnitude and number of impacts were recorded, and data were analyzed by total number of impacts, average magnitude of peak linear acceleration, cumulative linear exposure, and number of impacts ≥98g, an *a priori* high-magnitude impact threshold to be consistent with previous literature (21).

Instrumentation

The HITS is a wireless, instrumented-helmet system that provides real-time head impact kinematics for each impact and has been extensively described in the literature (8, 22, 23). Briefly, the HITS contains six single axis accelerometers inserted into each individual helmet (8, 22, 23). The data are transmitted to the Sideline Response System (Riddell Corp., Rosemont, IL, USA), where they are processed with the threshold set to record impacts above 10g. For any impact exceeding the threshold, 40ms of data (8ms pre-impact and 32ms post-impact) are recorded. The errors in peak linear acceleration magnitude between the HITS system and an instrumented head form ranged from 14.1%-29.5%, and the HITS recorded 91-96% of all impacts (24).

The SIM-G system is a wireless system that provides real-time head impact kinematics for each impact and has also been previously described in the literature (25, 26). The SIM-G accelerometers contain a high-g triaxial accelerometer, a low-g triaxial accelerometer, and a gyroscope (25, 26). The SIM-G accelerometers are embedded in elastic headbands and are positioned about the nuchal line. The data are transmitted to the custom software, where they are processed with the threshold set to record impacts above 10g. For any impact exceeding the threshold, 62ms of data (10ms pre-impact and 52ms post-impact) are recorded. The errors in peak linear acceleration magnitude between the SIM-G system and an instrumented head form ranged from 7.9%-20.6%, and the SIM-G recorded 99-100% of all impacts (24).

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Rotational acceleration was not included in the analyses because the SIM-G measures rotational velocity and derives rotational acceleration from this measure and the HITS does not directly measure rotational velocity or acceleration but estimates rotational acceleration based on data from an array of six linear accelerometers. Moreover, Cummiskey and colleagues report high errors in HITS peak rotational acceleration root mean square and mean absolute value of errors (24).

Tandem Gait (TG)

The TG is a test of dynamic balance, speed, and coordination. The narrow gait used in tandem gait is often considered a test of vestibular function during walking (27). The participants were instructed to stand barefoot behind the starting line with feet together and, in response to a verbal cue, they walked with alternating heel-to-toe gait, in a forward direction, along a 3-m long, 38-mm wide, line of sports tape as quickly as possible. Once the participants reached the end of the line, they completed a 180° turn and returned to the starting line with the same heel-to-toe gait pattern. To be considered a successful trial, the participants were required to complete the trial in ≤14 seconds without stepping off the line or separating the heel and toe. Each trial was timed using a handheld stopwatch and the fastest time of three trials was recorded. The TG is reliable (ICC: 0.97), sensitive (0.63) and specific (0.61) in detecting postural control impairments following concussion (28, 29).

Standard Assessment of Concussion (SAC)

The SAC is a mental status exam consisting of four sections: orientation, immediate memory, concentration, and delayed recall. It has a maximum total score of 30 points and each incorrect answer results in a loss of one point, so a higher score reflects better cognitive

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

performance. The SAC has moderate reliability (ICC: 0.34-0.39), sensitivity (0.95) and specificity (0.76) in detecting changes in mental status following concussion (30-32).

Balance Error Scoring System (BESS)

The BESS is commonly used for the evaluation of postural control following concussion (33). The BESS involves three different stances (double limb stance, single limb stance and tandem stance) on two surfaces (firm and foam). Each position is held for 20 seconds with the participants' eyes closed and hands on their hips. Errors are recorded by the administering clinician, which included moving the hands off the hips, opening the eyes, a step, stumble or fall, abduction or flexion of the hip beyond 30 degrees, lifting the forefoot or heel off the testing surface, and remaining out of the proper testing position for greater than 5 seconds and multiple errors occurring simultaneously are scored as a single error. Each stance has a maximum error score of ten and thus the total range for scoring was 0–60, where a lower score reflects "better" balance. Sensitivity for the BESS is low (0.34), but specificity is high (0.91-0.97) in detecting postural control impairments following concussion (33). The BESS has moderate reliability in a large collegiate cohort (ICC: 0.41) (32, 33).

King-Devick (KD)

The KD is a measurement of speed for rapid number naming and reading (34). It is designed to detect impaired eye movements and saccades (35). The test involves reading aloud a series of single-digit numbers from left to right on three test cards, each increasing in difficulty, as fast as possible. Timing starts when the participant begins to read each card and stops once he/she is finished. An error is counted if a number is incorrectly identified or if a number is skipped. Number of errors and overall time for all three cards are combined for each participant's score with a faster time (lower score) reflecting better performance. Each participant was tested

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

twice per session; the fastest time without errors was recorded as their PRE/POST score. The KD is reliable (ICC: 0.74) (32), sensitive (1.00) (36) and specific (0.94) in detecting changes in rapid number naming and reading following concussion (36).

Clinical Reaction Time (CRT)

The CRT ruler drop test is a simple measurement of reaction time (34). A clinician drops a 1.3m dowel rod marked in 0.5cm increments and embedded into a 256g rubber disc at random time intervals varying between two and five seconds and instructs the participants to grab the ruler as quickly as possible with their dominant hand once it drops. Each participant performed two practice trials followed by eight recorded trials with the forearm of their testing hand resting on a flat surface. The equation $d=1/2gt^2$ was used to convert the distance (in centimeters) that the ruler dropped to time (in milliseconds), whereby a faster time is better. The CRT has moderate reliability (ICC: 0.32) (32), and has 75% sensitivity and 68% specificity in detecting changes in RT following concussion (37).

Immediate Post-Concussion Assessment and Cognitive Testing (ImPACT)

The ImPACT is the most widely used computerized neuropsychological test for sport-related concussion baseline testing and evaluation (38). There are six test modules, which are used to evaluate attention span, working memory, sustained and selective attention time, response variability, non-verbal problem solving, and reaction time. From the six test modules, four composite scores are generated, including: 1) Verbal Memory, 2) Visual Memory, 3) Visual Motor Speed, and 4) Reaction Time. The ImPACT test was administered to each participant individually in a quiet room free from distraction. Combined test sensitivity is 0.81 and specificity is 0.89 in detecting neurocognitive impairments following concussion (39). Reliability varies (ICC: 0.34 - 0.72) depending on composite score measure (32).

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Statistical Analysis

Direct-entry linear regressions were used to determine if head impact kinematics and concussion history were predictors of change in clinical test battery performance from PRE to POST. The independent variables included total number of impacts, average magnitude of peak linear acceleration, cumulative linear acceleration, and number of impacts $\geq$98g, as well as concussion history for FB and WSOC independently. In addition, in a secondary analysis groups were combined, and direct-entry linear regressions were run with total number of impacts, number of impacts $\geq$98g, and concussion history only because WSOC and FB linear accelerations were recorded with different systems and HITS and SIM-G may not provide consistent head impact acceleration measures. Dependent variables included TG, SAC, BESS, KD, CRT, and ImPACT composite scores. In addition, associations between group (FB, WSOC) by time (PRE, POST) were calculated by a 2x2 repeated measures MANOVA. All analyses were completed using SPSS version 24.0 (SPSS Inc., Armonk, NY, USA). Level of significance for all comparisons was set *a priori* at $p<0.05$.

Results

WSOC data were recorded over 10 weeks and FB data were recorded over 12 weeks, which reflects the complete season for each sport. The FB (278.5±113.2) cohort sustained a greater number of RHI over the course of a season than the WSOC (142.9±118.8) cohort (Table 1).

Results from the direct-entry multiple regression analyses suggest that the overall associations between head impact exposure and both Visual Memory (R^2=0.670, F=6.487, p=0.002) and TG (R^2=0.636, F=3.841, p=0.029, Cohen's f^2=1.747) were significant for WSOC,

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

whereby those with greater exposure performed worse. In addition, the overall association between head impact exposure and KD (R^2=0.756, F=5.579, p=0.013, Cohen's f^2=3.098) was significant for FB, whereby those with greater exposure performed worse. No other regression analyses within or across groups were significant (Table 2).

The only predictors that made significant, unique contributions to the estimation of Visual Memory, TG, and KD was number of impacts ≥98g (Visual Memory: R^2=0.670, β=-0.537, p=0.005, Unstandardized B=-2.453; TG: R^2=0.636, β=0.511, p=0.036, Unstandardized B=1.736; KD: R^2=0.756, β=0.732, p=0.008, Unstandardized B=1.302) (Table 1, 3), whereby participants with more impacts ≥98g performed worse.

The multivariate effect showed that clinical test battery performance from PRE to POST differed across groups (Pillai's Trace=0.584, F=2.664, df[10, 19], p=0.032). Specifically, TG (F=12.858, p=0.001, η^2=0.315) and KD (F=4.263, p=0.048, η^2=0.132) improved more among WSOC than FB from PRE to POST (Table 4). Main effect for time was significant (Pillai's Trace=0.670, F=3.857, df[10, 19], p=0.006); however, the main effect for group was not significant (Pillai's Trace=0.510, F=1.977, df[10, 19], p=0.097), suggesting that both groups improved from PRE to POST, but there was no difference between FB and WSOC.

Discussion

The primary finding is that RHI in FB and WSOC does not result in clinically meaningful changes from PRE to POST on a multifaceted and multimodal neurological health assessment battery. The few significant PRE-POST change scores were related to the total number of high magnitude (≥98g) impacts sustained. Specifically, the regression model yielded significant results for Visual Memory and TG among WSOC and KD among FB, whereby those with

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

greater exposure to high linear acceleration impacts performed worse. Although it is unclear if these findings are meaningful or incidental, considering there were a total of 30 regression analyses performed and only 3 were significant, the linear fits between RHI and Visual Memory (R^2 = 0.670), TG (R^2 = 0.636), and KD (R^2 = 0.756) were high. The change scores that were associated with high magnitude impacts were those tests that evaluate visual/vestibular function and visual/vestibular dysfunction is common following concussion and RHI (40, 41). It is conceivable that these high magnitude impacts, although not resulting in acute signs and symptoms of concussion, may be associated with visual/vestibular impairments.

Our results are consistent with previous work indicating that FB do not demonstrate clinically meaningful changes from PRE to POST on measures of neurological function (8, 10), We also confirmed that changes in performance among FB were mostly independent of prior concussion history, and the total number and magnitude of impacts sustained over one season (8). It should be noted, however, that we used exclusively clinical concussion assessments whereas neuroimaging approaches have suggested subclinical changes in neurological function in response to RHI in FB, ice hockey, and soccer (11, 12, 14, 17, 18), though the clinical significance and the causation of these neuroimaging changes remains unknown. These data highlight the disconnect in the literature, whereby in the short-to-medium-term athletes do not present with clinically-observable behavioral neurological impairments (3, 4). However, it is possible that the subclinical changes that others have observed over the course of a season have a cumulative effect following years of repeated exposure that then become clinically-apparent later in life (42). Cognitive reserve is a theory based on more efficient utilization of brain networks or of enhanced ability to recruit alternate brain networks as needed that attempts to explain why there does not appear to be a direct relationship between the degree of brain pathology or brain

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

damage and the clinical manifestation of that damage (43, 44). It is possible that young, active, otherwise healthy collegiate student-athletes have sufficient cognitive reserves to mask the effects of RHI, but as these individuals age or suffer from head injury, these deficits become more apparent. Finally, there may be other variables, besides RHI, which are associated with later-life impairments as some individuals suffer from later-life complications, but many do not.

All clinical measures for participants in this study were generally within normative ranges for baseline testing in collegiate athletes (32). Across most clinical measures, participants improved from PRE to POST, which suggests that RHI are not associated with decreased neurological performance on clinical concussion assessments. However, WSOC had larger improvements than FB across the course of the season on TG ($p=.001$, $\eta^2=.315$) and KD ($p=.048$, $\eta^2=.132$) tests. It is possible that the FB group failing to improve as much as WSOC is a result of the greater exposure to RHI; however, without the presence of a male non-contact control group, it is difficult to interpret these findings. An improvement in common concussion assessment tools with repeated administration is not unexpected as these tests have known learning effects (32). It is also possible that FB have been exposed to these tests more frequently than WSOC, and therefore FB have less of a learning effect from repeat administration. The improvements observed among our cohort of athletes (Table 2) are within the expected change scores observed in previous work and these improvements may be attributed either to normal test-retest variance (32) or to a practice effect. We believe that there were no clinically meaningful changes observed across the season.

As expected, FB (278.5±113.2) sustain a greater number of RHI over the course of a season than WSOC (142.9±118.8). These data suggests that collision sports, such as football, have greater exposure to RHI than soccer (45). The number of impacts sustained by our FB

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

cohort is relatively low compared to prior research (8, 22, 23). We speculate that the relatively low number of impacts may be a result of increased awareness of the potential effects of RHI in sport, rule changes in NCAA participation, the level of play among our cohort, or even individual differences in athletes and teams style of play. These findings highlight the need to assess larger cohorts across several universities and levels of play. We chose not to compare the magnitude of head accelerations across sport because the data were collected with different systems. Although both systems have been validated and used extensively in the literature (22, 23, 25, 26), accelerometer systems affixed to the head in different ways with different processing algorithms may provide variable measures of head impact acceleration (24). Future research should explore using the same accelerometer system across a variety of sports to gain a better understanding of the differences in head impact magnitude.

This study was designed to investigate RHI in WSOC and FB, therefore the lack of a non-contact control group was a study limitation. We also acknowledge that we have made numerous comparisons and perhaps were able to find spurious statistical significance since we did not correct for multiple comparisons (i.e. Bonferroni correction would suggest 0.05/30 = 0.002). However, considering the implications of our findings (i.e. RHI do not result in many clinical changes), we chose a more liberal level of significance, $p = 0.05$, which would make it more likely to identify changes, and yet, we were still unable to identify many significant changes following RHI. We believe that providing effect sizes and Beta values further substantiate our findings. In addition, future work should employ the same accelerometer system across sports (e.g., SIM-G in FB players also wearing HITS) and incorporate a non-contact control group of similar activity level and matched by sex. The HITS system was only available in a specific Riddell helmet models, which limited the number of potential FB participants. The

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

high magnitude (>98g) threshold was selected to be consistent with previous work (18), but more research on what is considered a high-magnitude impact is warranted. Furthermore, following athletes over the course of multiple seasons in a longitudinal study would enhance the scope and translational application of this work. For example, we could then determine whether these observed impairments resolve before the next season or persist and compound in subsequent seasons. Finally, future research should incorporate not only clinical measures of neurological function, but also subclinical measures, such as fMRI, diffusion tensor imaging, genetic sequencing, and advanced postural control assessments in the same study.

In conclusion, we observed few significant PRE-POST clinical neurological changes among collegiate FB and WSOC athletes. These data suggest that RHI in sport does not result in clinically meaningful short-term changes on a multifaceted and multimodal concussion assessment battery. However, there may be subtle visual/vestibular impairments following RHI as observed by significant results for Visual Memory and TG among WSOC and KD among FB, whereby those with greater exposure to high linear acceleration impacts performed worse. These deficits were dependent on the sustained number of impacts ≥98g, suggesting that the number of high magnitude impacts may be more important than cumulative linear exposure.

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Acknowledgements

This publication was made possible, in part, with support from the Grand Alliance CARE Consortium, funded by the NCAA and the DoD. The USAMRAA, Ford Detrick, MD, USA, is the awarding and administering acquisition office. This work was supported by the Office of the Assistant Secretary of Defense for Health Affairs through the Psychological Health and Traumatic Brain Injury Program under Award No. W81XWH-14-2-0151. Opinions, interpretations, conclusions and recommendations are those of the authors and are not necessarily endorsed by the DoD (DHP funds).

Conflict of Interests

The authors have no conflicts of interest to disclose. The results of the present study do not constitute endorsement by ACSM. The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation.

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

References

1. Bailes JE, Petraglia AL, Omalu BI, et al. Role of subconcussion in repetitive mild traumatic brain injury: A review. *J Neurosurg*. 2013;119(5):1235-1245.

2. Montenigro PH, Alosco ML, Martin BM, et al. Cumulative head impact exposure predicts later-life depression, apathy, executive dysfunction, and cognitive impairment in former high school and college football players. *J Neurotrauma*. 2017;34(2):328-340.

3. Huber BR, Alosco ML, Stein TD, McKee AC. Potential long-term consequences of concussive and subconcussive injury. *Phys Med Rehabil Clin N Am*. 2016;27(2):503-511.

4. Hunter LE, Branch CA, Lipton ML. The neurobiological effects of repetitive head impacts in collision sports. *Neurobiol Dis*. 2018 Epub Jun 21. doi: 10.1016/j.nbd.2018.06.016.

5. Caplan B, Bogner J, Brenner L, et al. Subconcussive blows to the head: a formative review of short-term clinical outcomes. *J Head Trauma Rehabil*. 2016;31(3):159-66.

6. McAllister T, McCrea M. Long-term cognitive and neuropsychiatric consequences of repetitive concussion and head-impact exposure. *J Athl Train.* 2017;52(3):309-17.

7. Mainwaring L, Pennock KM, Mylabathula S, Alavie BZ. Subconcussive head impacts in sport: a systematic review of the evidence. *Int J Psychophysiol*. 2018 Epub Feb 3. doi: 10.1016/j.ijpsycho.2018.01.007.

8. Gysland SM, Mihalik JP, Register-Mihalik JK, et al. The relationship between subconcussive impacts and concussion history on clinical measures of neurologic function in collegiate football players. *Ann Biomed Eng*. 2012;40(1):14-22.

9. Miller JR, Adamson GJ, Pink MM, et al. Comparison of preseason, midseason, and postseason neurocognitive scores in uninjured collegiate football players. *Am J Sports Med*. 2007;35(8):1284-1288.

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

10. McAllister TW, Flashman LA, Maerlender A, et al. Cognitive effects of one season of head impacts in a cohort of collegiate contact sport athletes. *Neurology*. 2012;78(22):1777-1784.

11. Talavage TM, Nauman EA, Breedlove EL, et al. Functionally-detected cognitive impairment in high school football players without clinically-diagnosed concussion. *J Neurotrauma*. 2014;31(4):327-338.

12. Breedlove EL, Robinson M, Talavage TM, et al. Biomechanical correlates of symptomatic and asymptomatic neurophysiological impairment in high school football. *J Biomech*. 2012;45(7):1265-1272.

13. Mollayeva T, El-Khechen-Richandi G, Colantonio A. Sex & gender considerations in concussion research. *Concussion*. 2018;3(1):CNC51.

14. Sollmann N, Echlin PS, Schultz V, et al. Sex differences in white matter alterations following repetitive subconcussive head impacts in collegiate ice hockey players. *Neuroimage: Clinical*. 2018;17:642-649.

15. Lynall RC, Clark MD, Grand EE, et al. Head impact biomechanics in women's college soccer. *Med Sci Sports Exerc*. 2016;48(9):1772-1778.

16. Tarnutzer AA, Straumann D, Brugger P, et al. Persistent effects of playing football and associated (subconcussive) head trauma on brain structure and function: A systematic review of the literature. *Br J Sports Med*. 2017;51:1592-1604.

17. Stewart WF, Kim N, Ifrah C, et al. Heading frequency is more strongly related to cognitive performance than unintentional head impacts in amateur soccer players. *Front Neurol*. 2018 Epub April 24. doi: 10.3389/fneur.2018.00240

18. Lipton ML, Kim N, Zimmerman ME, et al. Soccer heading is associated with white matter microstructural and cognitive abnormalities. *Radiology*. 2013;268(3):850-857.

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

19. Broglio SP, McCrea M, McAllister T, et al. A national study on the effects of concussion in collegiate athletes and US military service academy members: The NCAA–DoD concussion assessment, research and education (CARE) consortium structure and methods. *Sports Med*. 2017;47(7):1437-1451.

20. McCrory P, Meeuwisse W, Dvorak J, et al. Consensus statement on concussion in sport—the 5th International Conference on Concussion in Sport held in Berlin, October 2016. *Br J Sports Med.* 2017;51:838-847.

21. Buckley TA, Oldham JR, Watson DJ, Murray NG, Munkasy BA, Evans KM. Repetitive Head Impacts in Football Do Not Impair Dynamic Postural Control. *Med Sci Sports Exerc*. 2018 Epub August 23. doi: 10.1249/MSS.0000000000001761.

22. Crisco JJ, Fiore R, Beckwith JG, et al. Frequency and location of head impact exposures in individual collegiate football players. *J Athl Train*. 2010;45(6):549-559.

23. Crisco JJ, Wilcox BJ, Beckwith JG, et al. Head impact exposure in collegiate football players. *J Biomech*. 2011;44(15):2673-2678.

24. Cummiskey B, Schiffmiller D, Talavage TM, et al. Reliability and accuracy of helmet-mounted and head-mounted devices used to measure head accelerations. *Proc. Inst. Mech. Eng. P*. 2017;231(2):144-153.

25. Lamond LC, Caccese JB, Buckley TA, et al. Linear acceleration in direct head contact across impact type, player position, and playing scenario in collegiate women's soccer players. *J Athl Train*. 2018;53(2):115-121.

26. Caccese JB, Lamond LC, Buckley TA, et al. Reducing purposeful headers from goal kicks and punts may reduce cumulative exposure to head acceleration. *Res Sports Med*. 2016;24(4):407-415.

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

27. Dozza M, Wall Iii C, Peterka RJ, Chiari L, Horak FB. Effects of practicing tandem gait with and without vibrotactile biofeedback in subjects with unilateral vestibular loss. *J Vestib Res*. 2007;17(4):195-204.

28. Oldham JR, DiFabio MS, Kaminski TW, et al. Normative tandem gait in collegiate student-athletes: Implications for clinical concussion assessment. *Sports Health*. 2017;9(4):305-311.

29. Oldham JR, Difabio MS, Kaminski TW, et al. Efficacy of tandem gait to identify impaired postural control after concussion. *Med Sci Sports Exerc*. 2018;50(6):1162-1168.

30. McCrea M. Standardized mental status assessment of sports concussion. *Clin J Sport Med*. 2001;11(3):176-181.

31. McCrea M. Standardized mental status testing on the sideline after sport-related concussion. *J Athl Train*. 2001;36(3):274.

32. Broglio SP, Katz BP, Zhao S, et al. Test-retest reliability and interpretation of common concussion assessment tools: Findings from the NCAA-DoD CARE consortium. *Sports Med*. 2018;48(5):1255-1268.

33. Buckley TA, Oldham JR, Caccese JB. Postural control deficits identify lingering post-concussion neurological deficits. *JSHS*. 2016;5(1):61-69.

34. Katz BP, Kudela M, Harezlak J, et al. Baseline performance of NCAA athletes on a concussion assessment battery: A report from the CARE consortium. *Sports Med*. 2018;48(80:1971-1985.

35. Galetta KM, Brandes LE, Maki K, et al. The King–Devick test and sports-related concussion: Study of a rapid visual screening tool in a collegiate cohort. *J Neurol Sci*. 2011;309(1):34-39.

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

36. King D, Gissane C, Hume PA, Flaws M. The King–Devick test was useful in management of concussion in amateur rugby union and rugby league in new zealand. *J Neurol Sci*. 2015;351(1):58-64.

37. Eckner JT, Kutcher JS, Broglio SP, Richardson JK. Effect of sport-related concussion on clinically measured simple reaction time. *Br J Sports Med*. 2014;48(2):112-118.

38. Kelly KC, Jordan EM, Joyner AB, et al. National collegiate athletic association division I athletic trainers' concussion-management practice patterns. *J Athl Train*. 2014;49(5):665-673.

39. Schatz P, Pardini JE, Lovell MR, et al. Sensitivity and specificity of the ImPACT test battery for concussion in athletes. *Arch Clin Neuropsychol*. 2006;21(1):91-99.

40. Master CL, Master SR, Wiebe DJ, et al. Vision and vestibular system dysfunction predicts prolonged concussion recovery in children. *Clin J Sport Med*. 2018;28(2):139-145.

41. Hwang S, Ma L, Kawata K, et al. Vestibular dysfunction after subconcussive head impact. *J Neurotrauma*. 2017;34(1):8-15.

42. Tremblay S, Pascual-Leone A, Théoret H. A review of the effects of physical activity and sports concussion on brain function and anatomy. *Int J Psychophysiol*. 2017; 132(A):167-175.

43. Stern Y. What is cognitive reserve? Theory and research application of the reserve concept. *J Int Neuropsychol Soc*. 2002;8(3):448-60.

44. Stern Y. Cognitive reserve. *Neuropsychologia*. 2009;47(10):2015-28.

45. Reynolds BB, Patrie J, Henry EJ, et al. Comparative analysis of head impact in contact and collision sports. *J Neurotrauma*. 2017;34(1):38-49.

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Table 1. Demographic and RHI kinematics information for women's soccer (WSOC) and men's football (FB) athletes. Data represent mean ± SD. The 95% confidence intervals are in parentheses. FB were taller (F=82.5,p<0.001) and heavier (F=116.2,p<0.001) than WSOC.

	WSOC	**FB**
N	23	15
Age (years)	19.7±1.2	19.5±1.0
Height (cm)	168.3±4.2	179.3±5.8
Weight (kg)	62.0±4.5	109.5±15.5
Concussion History	0.6±1.1 (0.1-1.0)	0.5±0.9 (0.0-1.0)
Position	Forward, N=6 Midfield, N=11 Defense, N=3 Goalkeeper, N=3	Offensive Lineman, N=4 Linebacker, N=3 Defensive Lineman, N=2 Tight End, N=1 Defensive Back, N=1 Wide Receiver, N=1 Punter, N=1 Kick Returner, N=1 Long Snapper, N=1
Number of Head Impacts	142.9±118.8 (86.9-189.3)	278.5±113.2 (215.8-341.3)
Mean Linear Acceleration (g)	16.1±3.6 (14.7-17.9)	23.8±8.0g (19.4-28.2)

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Cumulative Exposure (g)	2304.4±1866.7 (1432.8-3036.4)	6657.1±3747.8 (4581.7-8732.6)
Number of Impacts > 98g	0.2±0.4 (0.0-0.4)	1.1±1.0 (0.5-1.6)

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Table 2: Results of the regression model for clinical measures of neurologic function from PRE to POST. The change scores represent the average of individual differences POST-PRE.

	Preseason		**Postseason**		**Change**		**Regression Model**		
Men's Football									
Outcome Measure	*Mean*	*SD*	*Mean*	*SD*	*Mean*	*SD*	R^2	*F*	*p*
Tandem Gait	9.63	1.75	9.93	1.57	.17	1.60	0.277	0.536	0.745
BESS Total	11.91	5.88	8.74	4.35	-3.17	6.26	0.266	0.651	0.668
SAC Total	27.48	1.86	28.09	1.95	.61	2.50	0.308	0.800	0.576
CRT	19.92	3.66	21.55	4.89	1.64	4.48	0.324	0.864	0.540
KD	39.25	5.14	38.11	4.76	-1.14	3.42	0.756	5.579	0.013
ImPACT Composite Scores									
Verbal Memory	90.39	10.51	91.61	9.47	1.22	6.91	0.193	0.382	0.848
Visual Memory	77.39	15.44	80.57	13.43	3.17	13.19	0.600	2.400	0.130
Visual Motor Speed	44.43	6.25	43.06	6.35	-1.38	4.78	0.274	0.604	0.700
RT	.53	.04	.53	.05	.00	.04	0.495	1.568	0.272
Women's Soccer									
Outcome Measure	*Mean*	*SD*	*Mean*	*SD*	*Mean*	*SD*	R^2	*F*	*p*
Tandem Gait	10.94	2.10	9.02	1.79	-2.07	1.80	0.636	3.841	0.029
BESS Total	16.13	6.60	12.80	5.67	-3.33	5.33	0.263	1.142	0.379
SAC Total	27.47	2.17	27.47	2.03	.00	1.73	0.160	0.611	0.693
CRT	21.65	4.97	20.35	3.55	-1.30	4.98	0.176	0.682	0.644
KD	37.07	6.42	34.15	5.36	-2.92	3.46	0.197	0.787	0.574

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

ImPACT Composite Scores									
Verbal Memory	90.29	8.91	88.57	8.25	-1.71	7.56	0.199	0.793	0.570
Visual Memory	84.14	6.32	86.07	11.30	1.93	11.48	0.670	6.487	0.002
Visual Motor Speed	43.14	6.42	40.45	5.59	-2.68	3.19	0.074	0.254	0.931
RT	.56	.06	.59	.10	.03	.10	0.231	0.961	0.470
All Participants									
Outcome Measure	*Mean*	*SD*	*Mean*	*SD*	*Mean*	*SD*	R^2	*F*	*p*
Tandem Gait	10.16	1.98	9.54	1.70	-.80	2.00	0.012	0.105	0.956
BESS Total	13.58	6.44	10.34	5.24	-3.24	5.84	0.024	0.271	0.846
SAC Total	27.47	1.96	27.84	1.98	.37	2.22	0.007	0.074	0.973
CRT	20.60	4.25	21.08	4.40	.48	4.84	0.131	1.659	0.195
KD	38.39	5.70	36.55	5.31	-1.84	3.50	0.131	1.653	0.196
ImPACT Composite Scores									
Verbal Memory	90.35	9.81	90.46	9.03	.11	7.20	0.091	1.068	0.377
Visual Memory	79.95	13.08	82.65	12.80	2.70	12.42	0.085	0.989	0.410
Visual Motor Speed	43.94	6.25	42.07	6.13	-1.87	4.25	0.035	0.383	0.766
RT	.54	.05	.55	.08	.01	.07	0.078	0.908	0.448

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Table 3: Predictor variables for significant regression models identified in Table 2.

Group	**Variable**	**p**	***R^2***		**Total # Impacts**	**#>98 g**	**Cum. Expos.**	**Avg. Acc.**	**# Past Conc.**
Women's Soccer	*Visual Memory*	0.002	0.670	Beta	-1.648	-0.537	1.257	0.368	0.201
				Sig	0.165	0.005	0.299	0.337	0.271
Women's Soccer	*Tandem Gait*	0.029	0.636	Beta	0.877	0.511	-0.194	-0.239	0.023
				Sig	0.466	0.036	0.877	0.644	0.920
Men's Football	*KD*	0.013	0.756	Beta	1.683	0.732	-2.875	1.258	0.089
				Sig	0.153	0.008	0.057	0.146	0.676

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Table 4: Univariate results from repeated measures MANOVA for clinical measures of neurologic function from PRE to POST.

Outcome Measure	**F**	**Sig.**	**Partial Eta Squared**	**Observed Power**
Tandem Gait	12.858	.001	.315	.933
BESS Total	.216	.646	.008	.073
SAC Total	.533	.471	.019	.109
CRT	2.705	.111	.088	.355
KD	4.263	.048	.132	.513
ImPACT Composite Scores				
Verbal Memory	.585	.451	.020	.114
Visual Memory	.184	.671	.007	.070
Visual Motor Speed	.009	.924	.000	.051
RT	.986	.329	.034	.160

ACCEPTED

Copyright © 2019 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.